Microsoft Excel

Complete Study Notes: Basic to Advanced

Covers: Interface | Formulas | Functions | Charts | PivotTables | Data Analysis | Macros & VBA | Power Query | Advanced Techniques

Prepared for Placement & Academic Excellence

~ Prince yadav

SECTION 1: INTRODUCTION TO MS EXCEL

1. Introduction to MS Excel

Microsoft Excel is a spreadsheet application developed by Microsoft Corporation, part of the Microsoft Office Suite. It allows users to organize, format, calculate, and analyze data using a grid of rows and columns. Excel is one of the most widely used tools in business, academia, finance, data analysis, and everyday computing.

1.1 History & Versions

Year	Version	Key Feature
1985	Excel 1.0 (Mac)	First release for Apple Macintosh
1987	Excel 2.0 (Windows)	First version for Windows
1993	Excel 5.0	Introduction of VBA Macros
2003	Excel 2003	Improved XML support
2007	Excel 2007	Ribbon UI, .xlsx format (1M rows)
2010	Excel 2010	Sparklines, Slicers, PowerPivot
2013	Excel 2013	Flash Fill, Power Query
2016	Excel 2016	New chart types, Power BI integration
2019	Excel 2019	CONCAT, IFS, SWITCH functions
2021	Excel 2021	XLOOKUP, LET, dynamic arrays
2023+	Microsoft 365	LAMBDA, Python in Excel, Copilot AI

1.2 Key Uses of Excel

- Data Entry and Storage — Organizing structured data in rows and columns
- Mathematical Calculations — Using built-in functions and formulas
- Data Analysis — Sorting, filtering, pivot tables, statistical tools
- Data Visualization — Creating charts, graphs, dashboards
- Financial Modeling — Budgets, forecasts, P&L statements
- Automation — Macros and VBA for repetitive tasks
- Database Management — VLOOKUP, XLOOKUP, INDEX-MATCH
- Reporting — Professional formatted reports

1.3 Excel Interface Overview

When you open Excel, the screen is called the Workbook Window. Here are the key interface components:

Component	Description
Title Bar	Shows the name of the current workbook
Ribbon	Main menu with tabs: Home, Insert, Page Layout, Formulas, Data, Review, View
Quick Access Toolbar	Frequently used commands (Save, Undo, Redo) above the Ribbon
Name Box	Shows the address of the currently selected cell (e.g., A1)
Formula Bar	Displays and edits the content/formula of the active cell
Column Headers	Letters A, B, C... up to XFD (16,384 columns)
Row Headers	Numbers 1, 2, 3... up to 1,048,576 rows
Cell	Intersection of a row and column (e.g., B5)
Sheet Tab	Tabs at the bottom to switch between worksheets
Status Bar	Shows Sum, Count, Average of selected cells at the bottom
Scroll Bars	Horizontal and vertical bars to navigate the sheet

> 📝 **Note:** A single Excel workbook can contain multiple worksheets. Each worksheet has 1,048,576 rows and 16,384 columns — giving over 17 billion cells per sheet.

SECTION 2: EXCEL BASICS

2. Excel Basics

2.1 Workbook vs. Worksheet

Term	Definition	Example
Workbook	The entire Excel file (.xlsx)	Budget2024.xlsx
Worksheet (Sheet)	A single tab/page inside a workbook	Sheet1, Sales, Data
Cell	A single box at row-column intersection	A1, B5, Z100
Range	A group of cells	A1:D10, B2:B20
Active Cell	The currently selected cell	Highlighted with green border

2.2 Navigating Excel

Keyboard Navigation Shortcuts

Shortcut	Action
Arrow Keys	Move one cell in any direction
Ctrl + Arrow Key	Jump to last filled cell in that direction
Ctrl + Home	Go to cell A1
Ctrl + End	Go to last used cell
Ctrl + G or F5	Go To dialog box (jump to specific cell)
Page Up / Down	Move up/down one screen
Alt + Page Up/Down	Move left/right one screen
Ctrl + F	Find
Ctrl + H	Find & Replace
Tab	Move to next cell in the row
Shift + Tab	Move to previous cell in the row
Enter	Move down one row after input

2.3 Data Types in Excel

Excel recognizes different types of data and treats them differently:

Data Type	Description	Example	Alignment
Number	Numeric values for calculations	100, 3.14, -50	Right-aligned

Data Type	Description	Example	Alignment
Text (String)	Alphanumeric, cannot be calculated	Name, City, Code123	Left-aligned
Date/Time	Stored as serial numbers internally	01/01/2024, 14:30	Right-aligned
Boolean	TRUE or FALSE logical values	TRUE, FALSE	Center-aligned
Error	Formula error value	#DIV/0!, #REF!	Shows error text
Formula	Expression starting with =	=A1+B1, =SUM(A:A)	Depends on result

2.4 Entering and Editing Data

- Click a cell and start typing to enter data
- Press Enter to confirm and move down, Tab to confirm and move right
- Press Escape (Esc) to cancel data entry
- Double-click a cell or press F2 to edit its contents
- Press Delete to clear cell contents; press Backspace to delete during editing
- Use Fill Handle (small green square at bottom-right of cell) to copy or auto-fill data

> 💡 **Tip:** To enter data in multiple cells at once, select all cells first, type the value, then press Ctrl + Enter to fill all selected cells simultaneously.

2.5 Auto Fill and Flash Fill

Auto Fill

Auto Fill detects patterns and extends them:

- Drag the fill handle to copy values or extend a series
- Month names: Jan, Feb → drags to Mar, Apr, May...
- Numbers with pattern: 2, 4, 6 → extends to 8, 10, 12...
- Dates: 01/01/2024 → 02/01/2024, 03/01/2024...
- Right-click drag for options: Copy Cells, Fill Series, Fill Formatting Only

Flash Fill (Ctrl + E)

Flash Fill automatically fills values based on a pattern it detects from your input. Introduced in Excel 2013.

- Example: Column A has 'PRINCE SHARMA'. Type 'Prince' in B1, press Ctrl+E → Excel fills all first names
- Works for splitting, combining, reformatting text
- Very useful for cleaning imported data

2.6 Selecting Cells and Ranges

Selection	Method
Single cell	Click the cell
Range (contiguous)	Click and drag OR click first cell, Shift+click last
Multiple ranges	Select first range, hold Ctrl, select additional ranges
Entire row	Click the row number
Entire column	Click the column letter
Entire sheet	Click the triangle at row/column header intersection (Ctrl+A)
Named range	Type the name in the Name Box and press Enter

2.7 Copy, Cut, Paste Operations

- Ctrl + C — Copy selected cells
- Ctrl + X — Cut selected cells
- Ctrl + V — Paste
- Ctrl + Alt + V — Paste Special dialog

Paste Special Options

Paste Option	What It Does
All	Pastes everything (default)
Values	Pastes only the calculated values, not formulas
Formulas	Pastes only the formulas
Formats	Pastes only the formatting
Comments	Pastes only comments
Transpose	Switches rows and columns
Add / Subtract	Adds/subtracts clipboard value from destination
Column widths	Pastes the source column widths

> 📝 **Note:** 'Paste Values' is extremely important — it removes formulas and pastes only results, preventing errors when copying between sheets or workbooks.

2.8 Cell Formatting

Number Formats

Format	Example Input	Displayed As
General	12345.6	12345.6
Number	12345.6	12,345.60
Currency	12345.6	₹12,345.60 or $12,345.60

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Format	Example Input	Displayed As
Accounting	12345.6	₹ 12,345.60
Percentage	0.75	75%
Date	45292	01/01/2024
Time	0.5	12:00:00 PM
Text	007	007 (no leading zero removal)
Scientific	12345678	1.23E+07
Custom	Any	User-defined pattern

Font and Cell Formatting (Home Tab)

- Bold (Ctrl+B), Italic (Ctrl+I), Underline (Ctrl+U)
- Font size, font type, font color
- Fill Color (background color of cells)
- Text alignment: Left, Center, Right, Top, Middle, Bottom
- Wrap Text — allows content to wrap within a cell on multiple lines
- Merge & Center — merges selected cells into one (use sparingly)
- Borders — add lines around cells or ranges
- Indent — increase or decrease text indent within cell

2.9 Row and Column Operations

Operation	How To Do It
Insert Row	Right-click row number → Insert
Insert Column	Right-click column letter → Insert
Delete Row/Column	Right-click → Delete
Hide Row/Column	Right-click → Hide
Unhide Row/Column	Select surrounding rows/cols → Right-click → Unhide
Adjust Row Height	Drag bottom edge of row header OR right-click → Row Height
Adjust Column Width	Drag right edge of column header OR right-click → Column Width
AutoFit	Double-click edge of row/column header to auto-size to content
Freeze Panes	View tab → Freeze Panes (keeps headers visible while scrolling)

2.10 Undo, Redo, and Save

- Ctrl + Z — Undo last action (up to 100 actions by default)
- Ctrl + Y — Redo the undone action

- Ctrl + S — Save the workbook
- F12 — Save As (choose new name/location/format)
- Ctrl + W — Close the current workbook
- Ctrl + N — New workbook
- Ctrl + O — Open an existing workbook

> 💡 **Tip:** Set AutoSave interval: File → Options → Save → Save AutoRecover info every X minutes. This prevents data loss.

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3. Formulas and Cell References

A formula in Excel is an expression that calculates a value. All formulas begin with an equals sign (=). Formulas can include numbers, cell references, operators, and functions.

3.1 Basic Arithmetic Operators

Operator	Operation	Example	Result
+	Addition	=5+3	8
-	Subtraction	=10-4	6
*	Multiplication	=6*7	42
/	Division	=15/3	5
^	Exponentiation (Power)	=2^10	1024
%	Percentage	=50%	0.5
&	Text Concatenation	="Hello"&" "&"World"	Hello World

3.2 Order of Operations (BODMAS/PEMDAS)

Excel follows standard mathematical order of operations:

- Brackets/Parentheses — calculated first
- Exponentiation (^) — next
- Multiplication (*) and Division (/) — left to right
- Addition (+) and Subtraction (-) — left to right

> 📝 **Note:** Example: =2+3*4 gives 14 (not 20). Use =(2+3)*4 to get 20. Always use parentheses to control calculation order.

3.3 Cell References

Cell references tell Excel where to find the data to use in a formula. There are three types:

Reference Type	Syntax	Behavior When Copied	Use Case
Relative	A1	Adjusts based on copy direction	Most formulas
Absolute	$A$1	Never changes	Fixed constants (tax rate, GST %)
Mixed - Row Fixed	A$1	Column changes, row fixed	Fixed row reference

Reference Type	Syntax	Behavior When Copied	Use Case
Mixed - Col Fixed	$A1	Row changes, column fixed	Fixed column reference

Press F4 while editing a cell reference to cycle through reference types: A1 → $A$1 → A$1 → $A1 → A1

> 💡 **Tip:** Use absolute references ($) for values that must remain constant, like tax rates or conversion factors.

3.4 Referencing Other Sheets and Workbooks

`=Sheet2!B5`	Reference cell B5 in Sheet2

`='Sales Data'!C10`	Reference cell C10 in sheet named 'Sales Data' (quotes required when sheet name has spaces)

`=[Budget.xlsx]Sheet1!A1`	Reference cell A1 from another open workbook

3.5 Named Ranges

Named ranges let you refer to cells by a meaningful name instead of an address, making formulas easier to read and maintain.

Creating a Named Range

- Select the cell or range
- Click the Name Box (top-left, shows cell address)
- Type a name (no spaces; underscores allowed) and press Enter
- OR: Formulas tab → Define Name → Name Manager

Rules for Named Ranges

- Must start with a letter or underscore
- Cannot contain spaces (use underscore instead)
- Cannot be the same as a cell address (A1, B5, etc.)
- Case insensitive (TaxRate = taxrate)

`=SUM(SalesData)`	Sum all cells in a named range called 'SalesData'

`=Revenue - Cost`	Formula using named ranges for clarity

3.6 Common Formula Errors

Error	Cause	Fix
#DIV/0!	Division by zero or empty cell	Use IF() to check if denominator is 0
#VALUE!	Wrong data type in formula	Check data types; convert text to numbers
#REF!	Cell reference deleted or invalid	Check if referenced cells were deleted
#NAME?	Unrecognized formula name or text	Check function spelling; wrap text in quotes
#N/A	Value not found (e.g., VLOOKUP)	Use IFERROR() to handle gracefully
#NUM!	Invalid numeric value	Check for negative square roots or too large numbers
#NULL!	Incorrect range intersection	Check for missing : or comma in range
######	Column too narrow to display	Widen the column

> 💡 **Tip:** Use =IFERROR(formula, "Error Message") to display a custom message instead of error codes.

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4. Excel Functions

Functions are predefined formulas in Excel that perform specific calculations. Excel has 400+ built-in functions grouped into categories.

4.1 Mathematical & Statistical Functions

Function	Syntax	Description	Example
SUM	=SUM(range)	Adds all numbers in a range	=SUM(A1:A10)
AVERAGE	=AVERAGE(range)	Calculates arithmetic mean	=AVERAGE(B1:B50)
COUNT	=COUNT(range)	Counts cells with numbers	=COUNT(A1:A100)
COUNTA	=COUNTA(range)	Counts non-empty cells	=COUNTA(A1:A100)
COUNTBLANK	=COUNTBLANK(range)	Counts empty cells	=COUNTBLANK(A1:A50)
MAX	=MAX(range)	Returns largest value	=MAX(C1:C20)
MIN	=MIN(range)	Returns smallest value	=MIN(C1:C20)
MEDIAN	=MEDIAN(range)	Returns middle value	=MEDIAN(D1:D30)
MODE	=MODE(range)	Returns most frequent value	=MODE(E1:E40)
ROUND	=ROUND(num, digits)	Rounds to specified decimal places	=ROUND(3.14159,2)
ROUNDUP	=ROUNDUP(num, digits)	Always rounds up	=ROUNDUP(3.11,1)

Function	Syntax	Description	Example
ROUNDDOWN	=ROUNDDOWN(num, digits)	Always rounds down	=ROUNDDOWN(3.99,1)
CEILING	=CEILING(num, significance)	Rounds up to nearest multiple	=CEILING(11,5) = 15
FLOOR	=FLOOR(num, significance)	Rounds down to nearest multiple	=FLOOR(11,5) = 10
ABS	=ABS(num)	Returns absolute value	=ABS(-25) = 25
SQRT	=SQRT(num)	Square root	=SQRT(144) = 12
POWER	=POWER(num, power)	Raises number to a power	=POWER(2,8) = 256
MOD	=MOD(num, divisor)	Returns remainder after division	=MOD(17,5) = 2
INT	=INT(num)	Rounds down to nearest integer	=INT(5.9) = 5
TRUNC	=TRUNC(num, digits)	Truncates without rounding	=TRUNC(9.99) = 9
SUMIF	=SUMIF(range,criteria,sum_range)	Sum if condition met	=SUMIF(A:A,"Delhi",B:B)
SUMIFS	=SUMIFS(sum_range,criteria_range1,criteria1,...)	Sum with multiple conditions	=SUMIFS(C:C,A:A,"Delhi",B:B,">5000")
COUNTIF	=COUNTIF(range,criteria)	Count if condition met	=COUNTIF(A1:A100,">50")
COUNTIFS	=COUNTIFS(range1,crit1,range2,crit2,...)	Count with multiple conditions	=COUNTIFS(A:A,"Yes",B:B,">18")
AVERAGEIF	=AVERAGEIF(range,criteria,avg_range)	Average if condition met	=AVERAGEIF(A:A,"Sales",B:B)
STDEV	=STDEV(range)	Standard deviation (sample)	=STDEV(A1:A50)

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Function	Syntax	Description	Example
STDEVP	=STDEVP(range)	Standard deviation (population)	=STDEVP(A1:A50)
VAR	=VAR(range)	Variance (sample)	=VAR(B1:B20)
RANK	=RANK(num,ref,order)	Rank a number in a range	=RANK(A2,A:A,0)
LARGE	=LARGE(array,k)	Returns kth largest value	=LARGE(A1:A20,3)
SMALL	=SMALL(array,k)	Returns kth smallest value	=SMALL(A1:A20,2)
SUBTOTAL	=SUBTOTAL(function_num,range)	Subtotal ignoring hidden rows	=SUBTOTAL(9,B2:B100)

4.2 Text Functions

Function	Syntax	Description	Example
LEN	=LEN(text)	Returns number of characters	=LEN("Excel") = 5
LEFT	=LEFT(text,num_chars)	Extracts from left	=LEFT("Hello",3) = "Hel"
RIGHT	=RIGHT(text,num_chars)	Extracts from right	=RIGHT("Hello",3) = "llo"
MID	=MID(text,start,num_chars)	Extracts from middle	=MID("Hello",2,3) = "ell"
FIND	=FIND(find_text,within_text)	Position of text (case-sensitive)	=FIND("l","Hello") = 3
SEARCH	=SEARCH(find_text,within_text)	Position of text (not case-sensitive)	=SEARCH("L","Hello") = 3
UPPER	=UPPER(text)	Converts to uppercase	=UPPER("hello") = "HELLO"
LOWER	=LOWER(text)	Converts to lowercase	=LOWER("HELLO") = "hello"

Function	Syntax	Description	Example
PROPER	=PROPER(text)	Capitalizes first letter of each word	=PROPER("john doe") = "John Doe"
TRIM	=TRIM(text)	Removes extra spaces	=TRIM(" Hello ") = "Hello"
CLEAN	=CLEAN(text)	Removes non-printable characters	=CLEAN(A1)
SUBSTITUTE	=SUBSTITUTE(text,old,new)	Replaces specific text	=SUBSTITUTE(A1,"Old","New")
REPLACE	=REPLACE(text,start,chars,new)	Replaces by position	=REPLACE("Hello",1,3,"XYZ")
CONCATENATE	=CONCATENATE(t1,t2,...)	Joins text strings	=CONCATENATE(A1," ",B1)
CONCAT	=CONCAT(range)	Joins text (newer, range-friendly)	=CONCAT(A1:A5)
TEXTJOIN	=TEXTJOIN(delim,ignore,range)	Joins with delimiter	=TEXTJOIN(", ",TRUE,A1:A5)
TEXT	=TEXT(value,format)	Converts number to formatted text	=TEXT(45292,"DD/MM/YYYY")
VALUE	=VALUE(text)	Converts text number to actual number	=VALUE("123")
REPT	=REPT(text,times)	Repeats text N times	=REPT("*",5) = "*****"
CHAR	=CHAR(num)	Returns character for ASCII code	=CHAR(65) = "A"
CODE	=CODE(text)	Returns ASCII code of first char	=CODE("A") = 65
EXACT	=EXACT(text1,text2)	Case-sensitive comparison	=EXACT("Hello","hello") = FALSE

4.3 Logical Functions

Function	Syntax	Description	Example
IF	=IF(condition,true_val,false_val)	Returns value based on condition	=IF(A1>50,"Pass","Fail")
IFS	=IFS(cond1,val1,cond2,val2,...)	Multiple IF conditions (Excel 2019+)	=IFS(A1>=90,"A",A1>=80,"B",A1>=70,"C",TRUE,"D")
AND	=AND(cond1,cond2,...)	TRUE if ALL conditions are true	=AND(A1>10,B1<20)
OR	=OR(cond1,cond2,...)	TRUE if ANY condition is true	=OR(A1="Yes",B1="Yes")
NOT	=NOT(condition)	Reverses TRUE/FALSE	=NOT(A1>100)
IFERROR	=IFERROR(value,error_val)	Catches errors, shows alternative	=IFERROR(A1/B1,"Error")
IFNA	=IFNA(value,na_val)	Catches only #N/A errors	=IFNA(VLOOKUP(...),"Not Found")
SWITCH	=SWITCH(expr,val1,result1,...,default)	Match against values (like CASE)	=SWITCH(A1,1,"Jan",2,"Feb","Other")
XOR	=XOR(cond1,cond2,...)	TRUE if odd number of TRUE conditions	=XOR(A1>5,B1>5)
TRUE	=TRUE()	Returns logical TRUE	=TRUE()
FALSE	=FALSE()	Returns logical FALSE	=FALSE()

Nested IF Example

```
=IF(A1>=90,"A",IF(A1>=80,"B",IF(A1>=70,"C",IF(A1>=60,"D","F"))))
```

Grading formula using nested IFs — assign letter grade based on marks

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4.4 Lookup & Reference Functions

These are among the most critical functions for data analysis:

VLOOKUP — Vertical Lookup

`=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`	Searches for a value in the FIRST column of a table and returns a value from a specified column in the same row

Parameter	Description
lookup_value	The value to search for (e.g., Employee ID)
table_array	The table range to search in (e.g., $A$2:$E$100)
col_index_num	Column number to return (1=first col, 2=second, etc.)
range_lookup	FALSE = Exact match (recommended), TRUE = Approximate match

> 📝 **Note:** VLOOKUP limitations: Can only look left-to-right (lookup column must be leftmost), returns only one column, case-insensitive, returns first match only.

HLOOKUP — Horizontal Lookup

`=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])`	Same as VLOOKUP but searches horizontally across rows instead of columns

XLOOKUP — Modern Replacement (Excel 2021/365)

`=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`	More flexible lookup — can search any direction, return multiple columns, search from end

XLOOKUP Advantage	Description
Any direction	Can look left, right, up, or down
Default if not found	Built-in error handling parameter
Return multiple columns	Returns a range, not just one column
Wildcard support	Match mode 2 supports * and ? wildcards
Search from end	Search mode -1 finds last match

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XLOOKUP Advantage	Description
Binary search	Faster on sorted data (search mode 2 or -2)

INDEX and MATCH — The Power Combo

`=INDEX(return_range, MATCH(lookup_value, lookup_range, 0))`	INDEX returns a value at a position; MATCH finds the position. Together they replicate VLOOKUP with no column limitations

- MATCH(value, range, 0) — returns the row/column position of a value
- INDEX(array, row, [col]) — returns the value at that position
- Advantage: Works in any direction, more flexible than VLOOKUP

Other Reference Functions

Function	Syntax	Description
INDIRECT	=INDIRECT(ref_text)	Returns reference from a text string — dynamic ranges
OFFSET	=OFFSET(ref,rows,cols,[height],[width])	Returns a range offset from a starting cell
ADDRESS	=ADDRESS(row,col,abs_type)	Returns a cell address as text
ROW	=ROW([reference])	Returns the row number of a cell
COLUMN	=COLUMN([reference])	Returns the column number of a cell
ROWS	=ROWS(array)	Returns number of rows in a range
COLUMNS	=COLUMNS(array)	Returns number of columns in a range
CHOOSE	=CHOOSE(index,val1,val2,...)	Returns value from a list by index
MATCH	=MATCH(val,range,type)	Position of a value in a range
INDEX	=INDEX(array,row,[col])	Value at a specific position
TRANSPOSE	=TRANSPOSE(array)	Converts rows to columns and vice versa

4.5 Date and Time Functions

Function	Description	Example
TODAY()	Returns today's date (updates automatically)	=TODAY()
NOW()	Returns current date and time	=NOW()

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Function	Description	Example
DATE(y,m,d)	Creates a date from year, month, day	=DATE(2024,12,25)
YEAR(date)	Extracts year from a date	=YEAR(A1)
MONTH(date)	Extracts month number (1-12)	=MONTH(A1)
DAY(date)	Extracts day number (1-31)	=DAY(A1)
WEEKDAY(date,type)	Returns day of week (1=Sunday)	=WEEKDAY(A1,2)
WEEKNUM(date)	Returns week number of the year	=WEEKNUM(A1)
DATEDIF(start,end,unit)	Difference between dates (Y/M/D)	=DATEDIF(A1,B1,"Y")
DAYS(end,start)	Number of days between two dates	=DAYS(B1,A1)
NETWORKDAYS(start,end)	Working days between dates	=NETWORKDAYS(A1,B1)
WORKDAY(start,days)	Date after N working days	=WORKDAY(A1,30)
EOMONTH(start,months)	Last day of month N months away	=EOMONTH(A1,0)
EDATE(start,months)	Date N months from start date	=EDATE(A1,6)
TIME(h,m,s)	Creates a time value	=TIME(14,30,0)
HOUR(time)	Extracts hour	=HOUR(A1)
MINUTE(time)	Extracts minute	=MINUTE(A1)
TEXT(date,format)	Formats date as text	=TEXT(A1,"MMMM YYYY")

> 📝 **Note:** Dates in Excel are stored as serial numbers (1 = January 1, 1900). You can perform arithmetic with dates — e.g., =B1-A1 gives the number of days between two dates.

4.6 Dynamic Array Functions (Excel 365/2021+)

Dynamic array functions return multiple values that automatically spill into adjacent cells:

Function	Description	Example
FILTER	Filter a range based on conditions	=FILTER(A2:C100,B2:B100="Delhi")
SORT	Sorts a range or array	=SORT(A2:B50,2,-1)
SORTBY	Sorts by one or more criteria	=SORTBY(A2:C20,C2:C20,-1)
UNIQUE	Returns unique values from a range	=UNIQUE(A2:A100)
SEQUENCE	Generates a sequence of numbers	=SEQUENCE(10,1,1,1)
RANDARRAY	Generates random numbers	=RANDARRAY(5,3,1,100,TRUE)
TEXTSPLIT	Splits text by delimiter	=TEXTSPLIT(A1,",")
VSTACK / HSTACK	Combines arrays vertically/horizontally	=VSTACK(A1:B5,D1:E5)

Function	Description	Example
TAKE / DROP	Returns first/last N rows or columns	=TAKE(A1:C100,10)
TOCOL / TOROW	Reshapes range to single column/row	=TOCOL(A1:D5)
WRAPCOLS	Wraps a column into a 2D array	=WRAPCOLS(A1:A20,5)

> 💡 **Tip:** Use the # symbol after a spilled range's first cell to reference the entire spill. E.g., if FILTER result starts at A1, use A1# to reference all spilled results.

4.7 Financial Functions

Function	Description	Example Use
PMT(rate,nper,pv)	Calculates loan payment	Monthly EMI for a loan
PV(rate,nper,pmt)	Present value of investment	Current worth of future cash flows
FV(rate,nper,pmt,pv)	Future value of investment	Value of SIP after N years
RATE(nper,pmt,pv)	Interest rate per period	Find interest rate from loan terms
NPV(rate,values)	Net present value	Investment viability
IRR(values)	Internal rate of return	Project profitability
NPER(rate,pmt,pv)	Number of periods for loan	How long to pay off debt
SLN(cost,salvage,life)	Straight-line depreciation	Annual depreciation amount
DB(cost,salvage,life,period)	Declining balance depreciation	Asset depreciation schedule

5. Data Management and Analysis

5.1 Sorting Data

Sorting arranges data in a specified order. Excel can sort by single or multiple columns.

Single Column Sort

- Select any cell in the column you want to sort
- Data tab → Sort A to Z (ascending) or Sort Z to A (descending)
- For numbers: Smallest to Largest / Largest to Smallest
- For dates: Oldest to Newest / Newest to Oldest

Multi-Level Sort

- Data tab → Sort → Add Level
- Set primary sort column first, then secondary, tertiary...
- Example: Sort by Department (A-Z), then by Salary (descending) within each department

> 📝 **Note:** Always include headers when sorting! Check 'My data has headers' in the Sort dialog to prevent your header row from being sorted with data.

5.2 Filtering Data

AutoFilter

- Select data range → Data tab → Filter (or Ctrl + Shift + L)
- Drop-down arrows appear in each column header
- Click drop-down to filter by value, condition, or search
- Multiple filters can be active simultaneously
- Status bar shows count of visible rows

Filter Options

Filter Type	Options
Number Filters	Equals, Greater Than, Less Than, Between, Top 10, Above/Below Average
Text Filters	Equals, Contains, Begins With, Ends With, Does Not Contain
Date Filters	Today, Yesterday, This Week, This Month, This Year, Before, After, Between
Color Filters	Filter by cell color or font color

Filter Type	Options
Search	Type to search within filter dropdown

Advanced Filter

- Data tab → Advanced (under Sort & Filter group)
- Filter in place OR copy to another location
- Can extract unique records
- Criteria range: a separate area with column headers and filter conditions
- Use multiple rows in criteria range for OR conditions
- Use multiple columns in same row for AND conditions

5.3 Conditional Formatting

Conditional Formatting automatically applies formatting (colors, icons, data bars) to cells based on their values or formulas.

Types of Conditional Formatting

Type	Description	Use Case
Highlight Cell Rules	Color cells based on value comparison	Highlight scores > 90 in green
Top/Bottom Rules	Highlight top or bottom N values/percentages	Top 10 salespeople
Data Bars	Bar chart within cells proportional to value	Visual comparison of numbers
Color Scales	Gradient color based on value range	Heat map of sales by region
Icon Sets	Display icons (arrows, flags, stars) based on value	Status indicators
Formula-based Rules	Custom rule using any formula	Highlight rows where column D = "Pending"

Formula-Based Conditional Formatting Example

`=$D2="Overdue"`	Apply this rule to entire row (e.g., $A$2:$Z$1000) to highlight entire row when D column says 'Overdue'. Note: lock column with $, but not row.

> 💡 **Tip:** Use Conditional Formatting → Manage Rules to view, edit, delete, or reorder all rules in the current sheet.

5.4 Data Validation

Data Validation restricts what users can enter in cells, reducing data entry errors.

Setting Up Data Validation

- Select cells → Data tab → Data Validation
- Choose validation criteria type
- Set input message (tooltip when cell is selected)
- Set error alert (message shown when invalid data is entered)

Validation Type	Description	Example
Whole Number	Only integers within a range	1 to 100
Decimal	Numbers with decimals within range	0.0 to 5.0 (ratings)
List	Dropdown list of allowed values	Delhi, Mumbai, Chennai, Bangalore
Date	Dates within a range	Between 01/01/2024 and 31/12/2024
Time	Times within a range	09:00 to 18:00
Text Length	Restrict by character count	Between 8 and 15 characters
Custom	Use a formula to define rule	=ISNUMBER(FIND("@",A1)) for email

Dropdown List from a Range

- Select cells → Data Validation → List → Source: select range (e.g., $H$1:$H$10)
- Or type items directly: Delhi,Mumbai,Chennai (comma-separated)
- Using a range allows the dropdown to update automatically when you add new items

> 💡 **Tip:** Convert your list source to an Excel Table (Ctrl+T) to make the dropdown automatically include new entries added to the list.

5.5 Removing Duplicates

- Data tab → Remove Duplicates
- Choose which columns to check for duplicates
- Excel removes all duplicate rows and keeps only the first occurrence
- A dialog shows how many duplicates were removed

> 📝 **Note:** Always work on a copy of your data when removing duplicates. Consider using 'Remove Duplicates' after filtering rather than on raw data.

5.6 Text to Columns

Splits text in one column into multiple columns based on a delimiter or fixed width.

- Data tab → Text to Columns
- Delimited: split at comma, space, tab, semicolon, or custom character
- Fixed Width: split at specific character positions
- Set column data formats (Text, Date, General) in step 3

> 💡 **Tip:** Use Text to Columns to convert text-formatted dates or numbers imported from CSV files to proper Excel values.

5.7 Tables (Ctrl + T)

Excel Tables (not to be confused with general tables) are a powerful feature for managing structured data.

Creating a Table

- Select your data range → Ctrl + T (or Insert tab → Table)
- Check 'My table has headers' if first row is headers
- Excel assigns a name (Table1, Table2...) — rename it in Table Design tab

Benefits of Excel Tables

Feature	Benefit
Auto-expansion	Formulas and formatting automatically extend to new rows
Structured References	Use column names in formulas: =Table1[Sales] instead of C:C
Built-in Filters	AutoFilter applied automatically to headers
Total Row	One-click subtotals with function selection dropdown
Banded Rows	Alternating row colors for readability
Dynamic Charts	Charts based on tables update automatically with new data
Named Range Integration	Can use table names in VLOOKUP, SUMIF, etc.

SECTION 6: PIVOT TABLES & PIVOT CHARTS

6. PivotTables

PivotTables are one of Excel's most powerful features for summarizing, analyzing, and reporting large datasets. They allow you to quickly generate summaries, cross-tabulations, and aggregations without writing a single formula.

6.1 Creating a PivotTable

- Ensure your data is clean: no blank rows/columns, no merged cells, headers in row 1
- Select any cell in your dataset
- Insert tab → PivotTable → Choose location (New Worksheet recommended)
- The PivotTable Field List appears on the right side

6.2 PivotTable Field Areas

Area	Function	Example
Filters	Creates a report filter dropdown at the top	Filter by Year or Region
Columns	Field values become column headers	Months across the top
Rows	Field values become row labels	Products down the left
Values	Data to summarize (Sum, Count, Average, etc.)	Sum of Sales

> 📝 **Note:** Drag and drop fields between areas to instantly reshape your report. This is the core power of PivotTables.

6.3 Value Field Settings

Right-click a value in the PivotTable → Value Field Settings:

- Summarize by: Sum, Count, Average, Max, Min, Product, Count Numbers, StdDev, Var
- Show Values As: % of Grand Total, % of Row Total, % of Column Total, Running Total, Rank

6.4 PivotTable Tools

Tool	Description
Refresh	Updates PivotTable after source data changes (right-click → Refresh)
Change Data Source	Point PivotTable to a new or expanded range

Tool	Description
Group	Group dates by Year/Quarter/Month, or numbers into ranges
Ungroup	Remove grouping
Sort & Filter	Sort row/column labels or values within the PivotTable
Calculated Field	Add a custom formula column to the PivotTable
Show Details	Double-click a cell to see underlying data in a new sheet
PivotTable Styles	Apply built-in formatting styles (Design tab)
Subtotals & Grand Totals	Show or hide summary rows/columns

6.5 Slicers and Timelines

Slicers

- Visual filter buttons for PivotTables and Tables
- Insert tab → Slicer (or PivotTable Analyze tab → Insert Slicer)
- Click buttons to filter; Ctrl+Click for multiple selections
- One slicer can control multiple PivotTables (right-click → Report Connections)

Timelines

- Visual date filter — specific to date fields
- PivotTable Analyze tab → Insert Timeline
- Drag to select date range; switch between Year/Quarter/Month/Day view

6.6 Calculated Fields and Items

Calculated Field

- PivotTable Analyze → Fields, Items & Sets → Calculated Field
- Create a new field using other fields: = Sales - Cost
- The calculated field uses the same aggregation as the PivotTable

Calculated Item

- Creates a new item within a field based on existing items
- Example: Create a 'North Total' item that sums Delhi + Lucknow + Jaipur

6.7 PivotCharts

- Select any cell in PivotTable → PivotTable Analyze → PivotChart
- Chart dynamically updates when PivotTable filters change

- Has filter dropdowns built into the chart
- All standard chart types available
- Slicers also control PivotCharts linked to the same PivotTable

7. Charts and Data Visualization

Charts transform numerical data into visual representations, making trends, comparisons, and patterns immediately apparent. Excel offers 20+ chart types.

7.1 Creating Charts

- Select the data range (including headers)
- Insert tab → Charts group → Choose chart type
- Or press Alt + F1 for a quick embedded chart, or F11 for a chart on a new sheet
- Use 'Recommended Charts' (Insert → Recommended Charts) for AI-suggested best-fit charts

7.2 Chart Types and When to Use Them

Chart Type	Best For	Example Use
Column Chart	Comparing categories side by side	Monthly sales by product
Bar Chart	Same as column but horizontal — good for long category names	Survey results by response
Line Chart	Showing trends over time	Stock prices, temperature over months
Pie Chart	Showing proportions of a whole (use sparingly)	Market share by company
Doughnut Chart	Like pie but allows center label, can show multiple series	Budget allocation
Area Chart	Shows volume/magnitude of change over time	Cumulative revenue
Scatter (XY) Chart	Shows relationship/correlation between two variables	Height vs. weight, ad spend vs. revenue
Bubble Chart	Like scatter but adds 3rd dimension (bubble size)	Risk vs. return vs. investment size
Histogram	Shows frequency distribution	Score distribution of students
Box & Whisker	Shows distribution, quartiles, outliers	Salary distribution analysis
Waterfall Chart	Shows cumulative effect of sequential values	Profit & loss breakdown
Funnel Chart	Shows stages of a process	Sales pipeline conversion
Treemap	Hierarchical data by area/color	File sizes, market segments
Sunburst	Hierarchical data in concentric rings	Org structure, category hierarchy

Chart Type	Best For	Example Use
Combo Chart	Combines two chart types on same chart	Revenue (column) + Growth % (line)
Sparklines	Tiny inline charts within cells	Trend overview in a table

7.3 Chart Elements

Element	Description
Chart Title	Name displayed at the top of the chart
Plot Area	The area where data is plotted
Axes	X-axis (horizontal/category) and Y-axis (vertical/value)
Axis Titles	Labels explaining what each axis represents
Legend	Key showing which color/pattern represents which data series
Data Labels	Values shown directly on data points/bars
Gridlines	Reference lines in the plot area for reading values
Data Table	Table of values below the chart
Error Bars	Show statistical variability of data
Trendline	Line showing overall direction of data

> 💡 **Tip:** Click the + button next to a selected chart to quickly add/remove chart elements. Right-click any chart element to format it specifically.

7.4 Formatting Charts

- Right-click any chart element → Format [element]
- Use the Design tab to change chart layout, style, and color scheme
- Use the Format tab to format individual elements with shape/text styles
- Change chart type: Right-click chart area → Change Chart Type
- Move chart: Click chart → drag to new location; or Design tab → Move Chart
- Resize: Drag corner handles of the selected chart

7.5 Sparklines

Sparklines are mini-charts that fit inside a single cell, ideal for showing trends inline with tabular data.

- Insert tab → Sparklines → Line / Column / Win/Loss
- Select data range and choose location range (one cell per sparkline)
- Sparkline Tools (Design) tab appears when sparkline is selected
- Options: Show High/Low points, First/Last points, markers, style

8. Advanced Excel Features

8.1 What-If Analysis

What-If Analysis tools help model different scenarios to understand how changing inputs affects outputs.

Goal Seek

- Data tab → What-If Analysis → Goal Seek
- Set a formula cell to a specific value by changing a single input cell
- Example: Set cell B10 (Total Profit) to 100000 by changing B2 (Price per unit)
- Works backwards from desired result

Scenario Manager

- Data tab → What-If Analysis → Scenario Manager
- Create named scenarios with different input values
- Switch between scenarios to see different outcomes
- Create a Scenario Summary report on a new sheet
- Example: Best Case, Worst Case, Most Likely Case for a budget

Data Tables

- Data tab → What-If Analysis → Data Table
- One-variable table: shows how one input affects one output
- Two-variable table: shows how two inputs affect one output
- Creates an array of results automatically

8.2 Solver Add-In

- File → Options → Add-Ins → Manage: Excel Add-ins → Solver Add-in → Go → Check Solver
- Data tab → Solver (appears after enabling)
- More powerful than Goal Seek: optimize for max/min/specific value
- Can have multiple changing cells with constraints
- Uses linear programming / GRG Nonlinear / Evolutionary algorithms
- Example: Maximize profit (objective) by changing product quantities (variables) subject to resource constraints

8.3 Power Query (Get & Transform Data)

Power Query is Excel's ETL (Extract, Transform, Load) tool that allows you to connect to various data sources, transform data, and load it into Excel. Available in Data tab → Get & Transform Data.

Data Sources Power Query Can Connect To

- Excel files, CSV, Text files
- Databases: SQL Server, Access, Oracle, MySQL, PostgreSQL
- Web pages and APIs (JSON, XML)
- SharePoint, OneDrive, Azure
- PDF files, Folder (combine multiple files)

Common Transformations in Power Query

Transformation	Description
Remove Columns	Delete unnecessary columns
Remove Rows	Delete empty rows, errors, or based on conditions
Filter Rows	Keep only rows meeting criteria
Split Column	Split by delimiter or character count
Merge Columns	Combine multiple columns into one
Change Type	Set data types (text, number, date, etc.)
Replace Values	Find and replace specific values
Group By	Aggregate data by grouping columns
Unpivot	Convert wide format to long format (normalize data)
Pivot	Convert long format to wide format
Merge Queries	Join tables like SQL JOIN
Append Queries	Stack tables on top of each other (like UNION)
Add Custom Column	Create new column using M language formula
Conditional Column	Add column based on if/then logic
Index Column	Add sequential row numbers

Refreshing Power Query

- Data tab → Refresh All — reloads all queries from source
- Right-click query → Properties → set refresh schedule
- Ctrl + Alt + F5 — Refresh All keyboard shortcut

> 💡 **Tip:** Use Power Query when data comes from multiple sources, needs repeated cleaning, or is updated regularly. All transformation steps are recorded and replayable.

8.4 Power Pivot

Power Pivot is an advanced add-in for data modeling with large datasets, relationships between tables, and DAX (Data Analysis Expressions) measures.

~ Prince yadav

Key Concepts

- Data Model: Multiple related tables in memory (handles millions of rows)
- Relationships: Link tables by common key columns (like database foreign keys)
- DAX Measures: Calculated measures using DAX language (e.g., Total Revenue, YoY Growth)
- KPI: Key Performance Indicators built on DAX measures

Common DAX Functions

DAX Function	Description
SUM(table[column])	Sum of a column
CALCULATE(measure, filter)	Evaluate measure in modified filter context
FILTER(table, condition)	Return subset of table rows
ALL(table/column)	Remove all filters from a table or column
RELATED(column)	Get value from related table (like VLOOKUP)
DIVIDE(num, denom)	Safe division (no #DIV/0! error)
DISTINCTCOUNT(column)	Count unique values
DATEADD(dates, offset, interval)	Time intelligence — periods ago/ahead
SAMEPERIODLASTYEAR(dates)	Compare to same period last year
IF(condition, true, false)	Conditional logic in DAX

8.5 Macros & VBA

Macros automate repetitive tasks. VBA (Visual Basic for Applications) is the programming language behind Excel macros.

Recording a Macro

- Developer tab → Record Macro (enable Developer tab: File → Options → Customize Ribbon)
- Give the macro a name, optional shortcut key, and description
- Perform the actions you want to automate
- Click Stop Recording
- Run with: Developer tab → Macros, or the assigned shortcut key

VBA Editor (Alt + F11)

- Alt + F11 opens the Visual Basic Editor (VBE)
- Project Explorer (left): shows all open workbooks and their components
- Properties Window: shows properties of selected object
- Code Window: where you write and edit VBA code
- Insert → Module to add a new code module

VBA Basics

Concept	Syntax / Example
Sub procedure	Sub MacroName() ... End Sub
Function procedure	Function FuncName(args) As DataType ... End Function
Variables	Dim x As Integer, name As String
Data Types	Integer, Long, Double, String, Boolean, Date, Variant, Object
Assignment	x = 10 or name = "Prince"
Concatenation	MsgBox "Hello " & name
If-Then-Else	If x > 10 Then ... ElseIf ... Else ... End If
For Loop	For i = 1 To 10 ... Next i
For Each Loop	For Each cell In Range("A1:A10") ... Next cell
Do While Loop	Do While condition ... Loop
Select Case	Select Case x Case 1: ... Case 2: ... End Select
MsgBox	MsgBox "Message", vbOKCancel, "Title"
InputBox	result = InputBox("Enter value:", "Title")
With block	With Range("A1") .Value = 10 .Bold = True End With
Error handling	On Error GoTo ErrorHandler ... ErrorHandler: ...

Working with Excel Objects in VBA

Object	Example Code
Workbook	Workbooks("Budget.xlsx").Activate
Worksheet	Worksheets("Sheet1").Select or Sheets(1).Activate
Range	Range("A1:B10").Value = 0
Cell	Cells(1,1).Value = "Hello" (row, column)
Active Cell	ActiveCell.Value = "Test"
Selection	Selection.Font.Bold = True
Rows/Columns	Rows(3).Delete or Columns("B").Hidden = True
Last Row	lastRow = Cells(Rows.Count,1).End(xlUp).Row
Loop through cells	For Each c In Range("A1:A100") If c.Value > 50 Then ...
Chart	Charts.Add ActiveChart.ChartType = xlColumnClustered
UserForm	UserForm1.Show (for custom dialog boxes)

Macro Security

- File → Options → Trust Center → Trust Center Settings → Macro Settings
- Disable all macros (most secure) — macros blocked, notification shown

~ Prince yadav

- Disable with notification (recommended) — asks user before running
- Enable all macros (not recommended for general use)
- Save macro-enabled workbooks as .xlsm (not .xlsx)

8.6 Data Analysis ToolPak

An Excel add-in (File → Options → Add-ins → Analysis ToolPak) providing advanced statistical analysis:

Tool	Description
Descriptive Statistics	Mean, Median, Mode, Std Dev, Variance, Skewness, Kurtosis in one click
Histogram	Frequency distribution with chart
Correlation	Correlation matrix between multiple variables
Regression	Linear and multiple regression analysis
ANOVA	Analysis of Variance (one-way, two-way)
t-Test	Compare means of two groups
F-Test	Compare variances of two groups
Sampling	Draw random or periodic samples from data
Moving Average	Smooth data by calculating rolling averages
Exponential Smoothing	Forecast using weighted historical data
Random Number Generation	Generate sets of random numbers with distributions

8.7 Protection Features

Worksheet Protection

- Review tab → Protect Sheet
- Prevent editing of locked cells (cells are locked by default)
- Set password for unprotecting
- Allow specific actions: select cells, format cells, insert/delete rows, use PivotTables, etc.

Workbook Protection

- Review tab → Protect Workbook
- Prevent adding/deleting/renaming sheets, changing window size

File-Level Protection

- File → Info → Protect Workbook → Encrypt with Password
- Prevents opening the file without the password

> 📝 **Note:** If you forget a workbook password, Excel has no built-in recovery mechanism. Store passwords in a secure password manager.

Locking Specific Cells

- By default, all cells are locked but protection is not active
- To lock specific cells only: Select all cells → Ctrl+1 → Protection → Uncheck Locked
- Then select only the cells you want to lock → Check Locked
- Then Protect Sheet — only those selected cells will be protected

~ Prince yadav

9. Advanced Formulas and Techniques

9.1 Array Formulas

Array formulas perform calculations on multiple values simultaneously. In older Excel versions (pre-365), they were entered with Ctrl+Shift+Enter.

`=SUM(IF(A1:A100="Delhi",B1:B100,0))`	Array formula: Sum of B where A = Delhi. In older Excel, press Ctrl+Shift+Enter. In Excel 365, just press Enter (it's automatically dynamic).

`=MAX(IF(A1:A100="Manager",B1:B100))`	Find maximum salary of Managers — requires Ctrl+Shift+Enter in older Excel

9.2 LAMBDA Function (Excel 365)

LAMBDA allows you to create custom reusable functions in Excel without VBA.

`=LAMBDA(x, y, x^2 + 2*x*y + y^2)`	Define a function: $(x+y)^2$. Assign to a name via Name Manager.

- Create: Formulas → Name Manager → New → paste LAMBDA formula
- Name it (e.g., 'MySquare') → use it like: =MySquare(A1, B1)
- Can call itself recursively for complex calculations

9.3 LET Function (Excel 365)

LET assigns names to calculation results within a formula, improving readability and performance.

`=LET(revenue, SUM(B2:B100), cost, SUM(C2:C100), profit, revenue-cost, profit/revenue)`	Calculate profit margin using named variables inside the formula — each sub-calculation is only computed once

9.4 BYROW and BYCOL Functions

`=BYROW(A1:C10, LAMBDA(row, SUM(row)))`	Apply SUM to each row of a range — returns an array of row sums

`=BYCOL(A1:E5, LAMBDA(col, AVERAGE(col)))`	Apply AVERAGE to each column — returns an array of column averages

9.5 Working with Dates — Advanced

`=DATEDIF(B2,TODAY(),"Y")&" years, "&DATEDIF(B2,TODAY(),"YM")&" months"`	Calculate exact age from birthdate in years and months
`=WORKDAY(A1,30,HolidayList)`	Find the date 30 working days from A1, excluding holidays listed in HolidayList named range
`=TEXT(A1,"DDDD, DD MMMM YYYY")`	Format date as 'Monday, 01 January 2024'

9.6 Advanced Text Manipulation

`=TRIM(CLEAN(SUBSTITUTE(A1,CHAR(160)," ")))`	Remove all types of spaces including non-breaking spaces (common in data imported from web)
`=LEFT(A1,FIND(" ",A1)-1)`	Extract first word (everything before the first space)
`=MID(A1,FIND(" ",A1)+1,LEN(A1))`	Extract everything after the first space (last name or rest of text)
`=IFERROR(MID(A1,FIND("@",A1)+1,FIND(".",A1,FIND("@",A1))-FIND("@",A1)-1),"Invalid")`	Extract domain name from email address

9.7 Multi-Criteria Lookups

`=INDEX(C2:C100,MATCH(1,(A2:A100=F1)*(B2:B100=G1),0))`	Look up a value matching TWO criteria (enter with Ctrl+Shift+Enter in older Excel). Finds the row where column A = F1 AND column B = G1, then returns the corresponding value from column C.
`=XLOOKUP(F1&G1, A2:A100&B2:B100, C2:C100)`	XLOOKUP multi-criteria lookup by concatenating lookup values and arrays (Excel 365)

9.8 Dashboard Design Tips

- Use a dedicated 'Data' sheet and 'Dashboard' sheet — never show raw data on the dashboard
- Use PivotTables and GETPIVOTDATA for summary numbers on the dashboard
- Link KPI cards to cells using =Sheet1!A1 for dynamic values

~ Prince yadav

- Use named ranges for all data sources to make formulas readable
- Apply a consistent color palette (2-3 colors maximum)
- Use Slicers for interactive filtering — connect to multiple PivotTables
- Group related charts in the same visual area
- Protect the dashboard sheet to prevent accidental edits
- Hide gridlines: View tab → uncheck Gridlines
- Hide row/column headers: View tab → uncheck Headings
- Use shapes and text boxes for KPI tiles and section headers

~ Prince yadav

10. Essential Keyboard Shortcuts

10.1 General Shortcuts

Shortcut	Action
Ctrl + N	New Workbook
Ctrl + O	Open Workbook
Ctrl + S	Save
Ctrl + W	Close Workbook
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + P	Print
Ctrl + F	Find
Ctrl + H	Find & Replace
Ctrl + G or F5	Go To
F1	Help
Alt + F4	Exit Excel

10.2 Cell Editing Shortcuts

Shortcut	Action
F2	Edit active cell
Escape	Cancel cell entry
Delete	Clear cell contents
Ctrl + D	Fill Down
Ctrl + R	Fill Right
Ctrl + Enter	Fill all selected cells with current entry
Ctrl + ;	Insert current date
Ctrl + Shift + ;	Insert current time
Ctrl + '	Copy formula from cell above
Ctrl + `	Toggle Show Formulas mode
Ctrl + Shift + !	Apply Number format
Ctrl + Shift + $	Apply Currency format
Ctrl + Shift + %	Apply Percentage format
Ctrl + Shift + #	Apply Date format

Shortcut	Action
Ctrl + 1	Format Cells dialog
Ctrl + B	Bold
Ctrl + I	Italic
Ctrl + U	Underline
Alt + Enter	New line within a cell
F4	Repeat last action OR toggle cell reference type

10.3 Navigation Shortcuts

Shortcut	Action
Ctrl + Home	Go to A1
Ctrl + End	Go to last used cell
Ctrl + Arrow	Jump to edge of data region
Ctrl + Page Up/Down	Switch between sheets
Alt + Page Up/Down	Scroll one screen left/right
Ctrl + F6	Switch between open workbooks

10.4 Selection Shortcuts

Shortcut	Action
Ctrl + A	Select all cells (press twice for entire sheet)
Ctrl + Shift + Arrow	Extend selection to last data cell
Ctrl + Shift + End	Extend selection to last used cell
Ctrl + Shift + Home	Extend selection to A1
Shift + Space	Select entire row
Ctrl + Space	Select entire column
Ctrl + Shift + F3	Create names from selection
F8	Extend mode (extend selection without holding Shift)
Shift + F8	Add to selection

10.5 Formula Shortcuts

Shortcut	Action
=	Start a formula
Ctrl + Shift + Enter	Array formula (older Excel)
F9	Calculate formulas manually
Shift + F9	Calculate active sheet only

~ Prince yadav

Shortcut	Action
Ctrl + [	Trace dependents (show cells this formula uses)
Ctrl +]	Trace dependents (show cells using this cell)
Alt + M + P	Trace Precedents
Ctrl + `	Toggle display of formulas vs values
F3	Paste defined name into formula
Alt + =	AutoSum
Ctrl + Shift + A	Insert function arguments dialog

10.6 Data Shortcuts

Shortcut	Action
Ctrl + T	Create Table
Ctrl + Shift + L	Toggle AutoFilter
Alt + D + F + F	Toggle AutoFilter (older shortcut)
Alt + Down	Open dropdown in filtered column
Ctrl + E	Flash Fill
Alt + A + S + S	Sort dialog
Ctrl + Shift + F3	Create names from selection

10.7 PivotTable Shortcuts

Shortcut	Action
Alt + N + V	Insert PivotTable
Alt + J + T + R	Refresh PivotTable
Ctrl + Shift + *	Select entire PivotTable
Alt + J + T + F + F	Insert Calculated Field
Right-click → Expand/Collapse	Expand or collapse PivotTable groups

11. Exam & Interview Preparation

11.1 Frequently Asked Excel Interview Questions

Question	Key Points to Cover
What is the difference between VLOOKUP and XLOOKUP?	XLOOKUP works in any direction, has default if_not_found, returns ranges, no column number needed
Explain the difference between relative and absolute references	Relative adjusts when copied; Absolute ($) never changes; F4 cycles through types
What is a PivotTable? How do you create one?	Summarize/analyze data; Insert → PivotTable; rows/columns/values/filters areas
How do you handle errors in Excel formulas?	IFERROR, IFNA, wrap formula in error handler; specific error types and causes
What is the difference between COUNT, COUNTA, and COUNTBLANK?	COUNT = numbers only; COUNTA = non-empty (any type); COUNTBLANK = empty cells
Explain SUMIFS vs SUMIF	SUMIF = one condition; SUMIFS = multiple conditions; argument order differs
What is Power Query and when do you use it?	ETL tool for connecting, transforming, loading data; use for repeated data cleaning tasks
How do you use INDEX-MATCH?	INDEX returns value at position; MATCH finds position; together replace VLOOKUP with more flexibility
What is a macro? How do you record one?	Automates repetitive tasks; Developer tab → Record Macro; stored as VBA code
Explain conditional formatting	Auto-applies formatting based on rules/formulas; highlight cells, data bars, color scales, icon sets
How do you protect a worksheet?	Review → Protect Sheet; lock cells first; can allow specific actions; password optional
What is a named range and why use it?	Meaningful name for cell/range; makes formulas readable; defined in Formulas tab or Name Box

11.2 Quick Reference: Key Formula Patterns

Task	Formula
Sum if criteria met	=SUMIFS(sum_range, criteria_range, criteria)
Count unique values	=SUMPRODUCT(1/COUNTIF(A1:A100,A1:A100))
Running total	=SUM($B$2:B2) — anchor start, not end
Dynamic max without helper	=MAXIFS(B2:B100, A2:A100, "Criteria")
Extract N-th word	=TRIM(MID(SUBSTITUTE(A1," ",REPT(" ",100)),(N-1)*100+1,100))

~ Prince yadav

Task	Formula
Check if value in list	=ISNUMBER(MATCH(A1,List,0))
Age from birthdate	=DATEDIF(A1,TODAY(),"Y")
Last non-empty in column	=LOOKUP(2,1/(A1:A100<>""),A1:A100)
Percentage of total	=B2/SUM($B$2:$B$100)
Rank without ties	=SUMPRODUCT((B2>$B$2:$B$100)*1)+1
Days until deadline	=MAX(0,A1-TODAY())
Month name from date	=TEXT(A1,"MMMM")
Quarter from date	="Q"&INT((MONTH(A1)-1)/3)+1
Count distinct values	=ROWS(UNIQUE(A1:A100))
Find duplicates	=COUNTIF($A$1:$A1,A1)>1 — conditional format this

11.3 Best Practices for Professional Excel Work

- Always keep raw data on a separate, protected sheet — never edit source data
- Use tables (Ctrl+T) for all data ranges to ensure formulas expand automatically
- Document your work: add comments, use descriptive sheet names and named ranges
- Avoid hardcoding values in formulas — store constants in dedicated cells
- Use color coding for input cells (yellow), formula cells (blue/gray), output (green)
- Audit formulas regularly: Formulas tab → Trace Precedents / Dependents / Evaluate Formula
- Test edge cases: empty cells, zero values, text in number columns, dates out of range
- Avoid excessive volatile functions (TODAY(), NOW(), INDIRECT(), RAND()) — they recalculate every time
- Keep formulas simple — break complex formulas into multiple columns if needed
- Back up your work; use version names (Budget_v1.xlsx, Budget_Final.xlsx)
- Use IFERROR or IFNA consistently to handle formula errors gracefully
- Never merge cells in data tables — it breaks sorting, filtering, and PivotTables
- Use consistent date formats across the workbook
- Keep column headers concise and avoid special characters
- Test your dashboard/report on a fresh dataset before presenting

12. Glossary of Key Terms

Term	Definition
Absolute Reference	Cell reference that does not change when formula is copied ($A$1)
Array Formula	Formula that performs calculations on multiple values simultaneously
AutoFill	Feature that automatically extends patterns (dates, numbers, months) by dragging fill handle
AutoFilter	Filter feature that adds dropdown arrows to column headers
Cell	Intersection of a row and column in a worksheet (e.g., B5)
Conditional Formatting	Automatically applies formatting based on cell values or formulas
DAX	Data Analysis Expressions — formula language used in Power Pivot
Data Validation	Restricts the type of data that can be entered in a cell
Dynamic Array	Formulas that return multiple values and automatically spill into adjacent cells
Fill Handle	Small green square at bottom-right of selected cell for AutoFill
Flash Fill	Automatically fills data based on a pattern (Ctrl+E)
Formula Bar	Bar below the Ribbon showing contents of active cell
Goal Seek	What-If Analysis tool that finds input needed to achieve a target result
LAMBDA	Excel 365 function for creating custom reusable functions
Macro	Recorded or written set of instructions to automate tasks
Name Box	Box showing cell address or named range at top-left of worksheet
Named Range	User-defined name assigned to a cell or range of cells
PivotChart	Chart linked to a PivotTable that updates dynamically with filters
PivotTable	Tool to summarize and analyze data from a large dataset interactively

Term	Definition
Power Pivot	Excel add-in for data modeling with large datasets and DAX measures
Power Query	Excel's ETL tool for connecting to, transforming, and loading data
Range	A group of cells (e.g., A1:D10)
Relative Reference	Cell reference that adjusts when the formula is copied (A1)
Ribbon	The tabbed toolbar at the top of Excel with all commands
Slicer	Visual filter button panel for PivotTables and Tables
Solver	Add-in for optimization problems with multiple variables and constraints
Sparkline	Tiny chart that fits inside a single cell
Structured Reference	Table column names used in formulas: =Table1[Sales]
Timeline	Visual date filter tool for PivotTables
VBA	Visual Basic for Applications — programming language for Excel macros
VLOOKUP	Function to look up a value in a table and return related data
Volatile Function	Functions that recalculate every time Excel recalculates (TODAY, RAND, INDIRECT)
Workbook	The entire Excel file (.xlsx, .xlsm)
Worksheet	A single tab/sheet within a workbook
XLOOKUP	Modern, flexible replacement for VLOOKUP (Excel 2021/365)
XY Chart	Scatter chart showing correlation between two numerical variables

~ Prince yadav